Xiaochuan Li

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Research interests

Learning and representation of linguistic meaning; semantic and pragmatic inference; controlled language-model experiments and causal interpretability.

Education

Oct. 2025 – exp. 2027	**University of Technology Nuremberg**, Germany *M.Sc. Human and Artificial Intelligence* Coursework: deep learning, interpretability of language models, psychology and empirical study design, cognitive neuroscience.
Oct. 2024 – Jul. 2025	**University of Marburg**, Germany *German for graduate study*
Sep. 2020 – Jun. 2024	**Shenzhen University**, China *B.A. Philosophy* Coursework: formal and modal logic, philosophy of language.

Research experience

Ongoing	**StemmaLM** *Independent infrastructure for language-model lineages* Trained 60+ language models (up to 98M parameters) from scratch on child-scale corpora and controlled variants of them. Built the corpora, the training pipeline on an HPC cluster, and a public model registry. stemmalm.github.io
Oct. 2026	**What can be learned from vagueness** *Independent project* Tested whether disagreement between speakers lets a learner recover the order of categories from text alone, with a formal result and transformers and LSTMs trained from scratch. Preprint.
Apr. 2026	**Moral framing and dilemma choices in language models** *Project with the German Research Center for Artificial Intelligence (DFKI)* Led the utilitarian-framing branch: how moral framing changes dilemma choices in 7B and 14B models, studied with controlled prompts and activation patching. Manuscript in preparation.
Ongoing	**Modal language development in preschoolers** *M.Sc. thesis, Max Planck Institute for Human Cognitive and Brain Sciences* Behavioural study of how preschoolers acquire modal words such as *might* and *must*. In design.

Feb. 2026	**How the first information source shapes verification: a between-subjects experiment** *Course experiment (group of three), M.Sc. Human and AI* Participants checked general-knowledge answers with ChatGPT or Wikipedia first, assigned at random; from pilot (N = 8) and revision to online data collection on Pavlovia (49 participants, 47 analysed) and analysis in JASP. Co-designed the study, ran the analysis and wrote the report.
May 2024	**Leibniz's logical system and characteristica universalis** *B.A. thesis, Shenzhen University* Leibniz's logic and his proposal for a universal symbolic language in which reasoning becomes calculation.

Preprints

Li, X. (2026). *What can be learned from vagueness* [Preprint]. Zenodo. https://doi.org/10.5281/zenodo.23677586

Employment

Apr. 2026 – present	**University of Technology Nuremberg**, Germany *IT Service Desk (part-time)* Student IT support; basic HPC user administration (resource allocation, access rights).
Sep. – Dec. 2021	**Shenzhen Learning Disorders Association**, China *Internship · Learning assessment at Yucai No. 2 Primary School* Ran reading assessments with primary-school children referred for learning difficulties.

Skills and languages

Training	Small transformers and LSTMs trained from scratch; fine-tuning, LoRA
Interpretability	Activation patching, low-rank analysis, probing, steering
Evaluation	Controlled behavioural evaluation, minimal pairs
Programming	Python (PyTorch, Hugging Face Transformers, pandas, scikit-learn), Git, LaTeX, Slurm on an HPC GPU cluster
Human data	Experimental design, PsychoPy/Pavlovia, JASP
Languages	Mandarin Chinese (native), German (C1), English (C1), Japanese (learning)